Chest X-Ray Pneumonia Classification Using Interpretable Features and Machine Learning from Scratch

Ravichandra Shinde

Abstract - This project studies binary classification of NORMAL and PNEUMONIA chest radiographs using explicitly implemented classical machine learning. An audit of 5,856 readable images links filename-derived patient proxies and exact decoded-pixel hashes to reduce known overlap. After exclusions, 4,753 development images and 618 test images remain. Each grayscale image is resized with aspect-preserving padding to 64 by 64 pixels and represented by 149 regional and global intensity features. A majority baseline, Gaussian Naive Bayes and two regularization settings of logistic regression are compared using three grouped development folds with training-only standardization. Logistic regression with regularization 0.01 achieves the highest mean development balanced accuracy, 95.38%. At a fixed threshold of 0.5, the held-out test achieves 83.50% accuracy, 79.06% balanced accuracy, 96.64% recall, 61.47% specificity and 0.9240 ROC-AUC. A fixed-setting ablation favors regional features over global summaries alone. The generalization gap, proxy grouping and modest specificity limit interpretation. The work provides a reproducible educational benchmark and does not establish clinical readiness.

Index Terms - Chest radiographs, pneumonia classification, logistic regression, feature engineering, grouped validation.

I. INTRODUCTION

Chest radiographs provide a well-established setting for medical image classification research [1]. This project asks a narrower educational question: can a small set of interpretable regional intensity features, paired with transparent classical models, distinguish the supplied NORMAL and PNEUMONIA annotations under a documented validation protocol? The target is the dataset label. It is not a prospective diagnosis or an assessment of treatment need.

The contribution is an end-to-end implementation that makes the statistical operations inspectable. Image auditing, proxy-group assignment, feature extraction, standardization, Gaussian Naive Bayes, logistic optimization, model selection and evaluation metrics are implemented directly. NumPy supports arrays and arithmetic, Pillow decodes and resizes images, and Matplotlib produces plots. No machine-learning estimator, optimizer or library metric is used.

The experiment emphasizes the distinction between detecting annotated pneumonia and correctly rejecting normal images. Accuracy alone can conceal errors when labels are imbalanced. Balanced accuracy therefore controls development selection, while recall, specificity, precision, F1 and ROC-AUC describe the final result. All numerical claims in this report refer to the retained notebook experiment and its saved local outputs.

II. RELATED WORK

Kermany et al. study deep learning for medical image classification and supply the research context for this dataset [1]. Their results do not directly benchmark the present method because the representations, models and audited partitions differ. The present work uses coarse hand-engineered summaries rather than a learned neural representation.

Ng and Jordan compare logistic regression and Naive Bayes as discriminative and generative classifiers [2]. That contrast motivates including both models rather than assuming one must always dominate. General model-assessment principles, including regularization and validation, follow the framework described by Hastie, Tibshirani and Friedman [3].

Zech et al. document variation in pneumonia-model generalization across radiograph sources [4]. Their findings motivate caution about acquisition differences and external validity. The present internal benchmark cannot establish performance at a new hospital, and its proxy grouping cannot replace verified patient identifiers.

III. DATASET AND AUDIT

A. Source and original partitions

The input is the Kaggle Chest X-Ray Images (Pneumonia) dataset [5], organized into train, val and test folders with NORMAL and PNEUMONIA subfolders. The audit decodes exactly one canonical tree and ignores the nested duplicate copy present in the download. All 5,856 images in that tree decode successfully. Clinical age, sex, site and verified patient identifiers are unavailable in the project metadata and are not imputed.

TABLE 1. Original image counts before audit exclusions.

Split	NORMAL	PNEUMONIA	Total
Train	1,341	3,875	5,216
Validation	8	8	16
Test	234	390	624
Total	1,583	4,273	5,856

B. Overlap and duplicate policy

A SHA-256 digest of each decoded grayscale array, including its shape, identifies exact pixel duplicates. A filename parser derives conservative patient proxies. Matching hashes or proxies connect images into groups, including transitive links. Development images whose group touches the test set are excluded before model fitting. Label-conflicting duplicate pixels would be excluded, and one remaining exact copy is retained. The original radiographs are not modified.

The audit excludes 457 development images for proxy-group overlap with test and 28 further exact duplicates, including six test images. Thus 485 images are excluded overall. Training and the 16-image supplied validation folder form development because that tiny validation set is unsuitable as the sole selection set. The retained development set contains 1,348 NORMAL and 3,405 PNEUMONIA images. The retained test contains 231 NORMAL and 387 PNEUMONIA images.

These rules reduce identifiable leakage but do not prove patient independence. Filenames can collide, repeat patients may not share a recognized naming pattern, and exact hashes miss near duplicates. Test size, quality and exposure summaries are descriptive audits. They do not select features, regularization or the classification threshold.

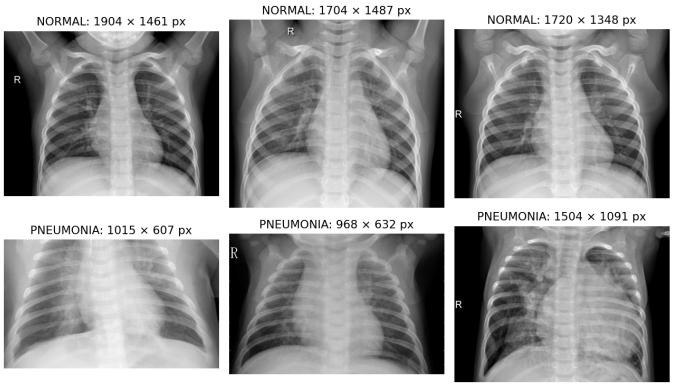


Fig. 1. Seeded development examples from the notebook. Labels are inherited annotations and do not independently verify diagnosis.

IV. FEATURE REPRESENTATION AND MODELS

A. Image preprocessing and 149 features

Each image is converted to grayscale and padded to 64 by 64 pixels with bilinear resizing while preserving aspect ratio. Pixel values are divided by 255. An 8 by 8 spatial grid then yields 64 block means and 64 block standard deviations. A 16-bin normalized intensity histogram and five global summaries complete the representation: image mean, standard deviation, median, mean absolute vertical pixel difference and mean absolute horizontal pixel difference.

Feature dimension: 64 + 64 + 16 + 5 = 149.

The regional summaries retain coarse spatial information with a low computational cost. Original dimensions, filenames, groups and split labels are never predictors. However, padding, markers and exposure may still influence pixel-derived summaries. The method does not establish that its features isolate pneumonia-specific pathology. Raw pixels, texture descriptors and learned features remain possible alternatives, with different complexity and information tradeoffs.

B. Training-only standardization

Division by 255 is fixed normalization. Standardization instead estimates a mean and standard deviation for each feature from the current training rows only. Each cross-validation fit computes its own scaler, and the final fit uses all development rows. A constant feature uses divisor one to keep transformed values finite.

z_ij = (x_ij - training_mean_j) / training_std_j; use divisor 1 for a constant feature.

Standardization gives differently scaled intensity summaries a more consistent numerical influence during gradient descent and L2 regularization. Fitting it before splitting would expose validation or test statistics to training. The notebook therefore fits preprocessing inside every development fold.

C. Majority baseline and Gaussian Naive Bayes

The majority model always predicts the most common training class. Its balanced accuracy is 0.5 when both classes occur, making it a useful check against misleading overall accuracy. Gaussian Naive Bayes estimates a separate mean and variance for each feature and class, assumes conditional independence and uses equal class priors. A variance floor of 0.001 in standardized squared units prevents singular densities.

Gaussian NB: log score(c) = log prior(c) - 0.5 sum_j [log(2 pi variance_cj) + (z_j - mean_cj)^2 / variance_cj].

Adjacent regional features are correlated, so the Naive Bayes independence assumption is approximate. Equal priors give equal class importance rather than reproducing development prevalence. These modeling choices can affect probability interpretation and calibration.

D. Weighted logistic regression

Logistic regression models a score using a sigmoid of a linear function of standardized features. A prediction is PNEUMONIA when the score is at least 0.5. Inverse-frequency weights give each class equal total influence within the current training fold. The objective combines weighted binary log-loss with an L2 penalty on coefficients. The bias is not penalized.

p_i = sigmoid(w^T z_i + b); class weight a_i = n / (2 n_yi).

J = mean[a_i (log(1 + exp(s_i)) - y_i s_i)] + (lambda / 2) ||w||^2.

*Gradient_w = Z^T [a * (p-y)] / n + lambda w; gradient_b = mean[a * (p-y)].*

Full-batch gradient descent begins with zero coefficients and bias and an initial step size of 0.2. Backtracking halves the step whenever it increases the objective. Optimization stops after at most 1,000 iterations or when absolute objective improvement falls below 0.00000001. The loss history remains inspectable. Weighted training changes the score interpretation, and the output is not established as a calibrated clinical probability.

V. EXPERIMENTAL PROTOCOL

All candidates use the same three approximately stratified group folds with seed 42. The explicit assignment algorithm shuffles groups, sorts larger groups first and assigns each group to the fold with the fewest samples of its majority class. This keeps groups intact but does not guarantee optimal stratification. The candidate set contains the majority model, Gaussian Naive Bayes and logistic regression with regularization 0.01 or 0.1.

Mean held-out fold balanced accuracy selects the model. Ties retain the first declared candidate. Scaling and class weights use training rows separately in each fit. Following selection, the scaler and winning model are refitted on all development rows. The final test uses the predeclared threshold of 0.5. Test outcomes never revise the winner, regularization, representation or threshold.

PNEUMONIA is the positive class. Accuracy measures correct predictions over all test rows. Recall is TP/(TP+FN), specificity is TN/(TN+FP), and balanced accuracy averages those two rates. Precision is TP/(TP+FP), and F1 is the harmonic mean of precision and recall. ROC-AUC uses trapezoidal integration with tied scores grouped together. Hyperparameter selection makes the reported development scores selection-biased rather than unbiased nested-cross-validation estimates.

Uncertainty intervals use 500 percentile bootstrap repetitions over retained test proxy groups with seed 42. Each repetition samples groups with replacement and includes their images. Repetitions without both labels are skipped. The intervals condition on the fitted model and proxy groups. They exclude training, selection and external-population uncertainty.

VI. RESULTS

A. Development comparison

TABLE 2. Development balanced accuracy on identical grouped folds. SD is descriptive fold variation in percentage points.

Candidate	Mean (%)	SD (pp)
Majority	50.00	0.00
Gaussian NB	84.21	1.50
Logistic, lambda=0.01	95.38	0.77
Logistic, lambda=0.1	93.17	0.84

Logistic regression with regularization 0.01 obtains the highest mean development balanced accuracy, 95.38%, compared with 84.21% for Gaussian Naive Bayes and 50.00% for the majority baseline. The stronger regularization candidate reaches 93.17%. Training balanced accuracy for the selected setting averages 96.05% across folds. The three validation scores share overlapping training data, so their standard deviation does not support a claim of independent-trial significance.

B. Held-out evaluation

TABLE 3. Test performance of the selected model on 618 images.

Metric	Value
Accuracy	83.50%
Balanced accuracy	79.06%
Precision	80.78%
Recall / sensitivity	96.64%
Specificity	61.47%
F1	0.8800
ROC-AUC	0.9240

TABLE 4. Confusion counts at the fixed threshold 0.5.

Actual / predicted	NORMAL	PNEUMONIA
NORMAL	142	89
PNEUMONIA	13	374

The model identifies 374 of 387 annotated pneumonia images and correctly rejects 142 of 231 normal images. Its 13 false negatives accompany 89 false positives. The difference between mean development and test balanced accuracy is 16.32 percentage points. High recall therefore coexists with modest specificity and a substantial generalization gap. ROC-AUC describes ranking quality and does not by itself establish a suitable operating point.

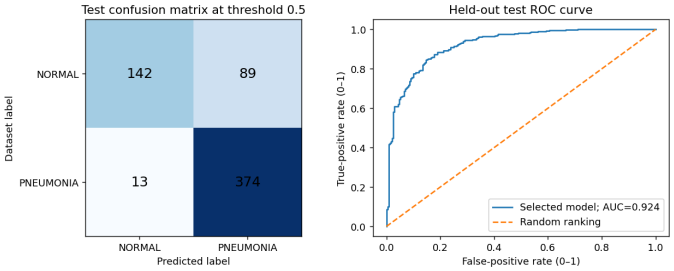


Fig. 2. Saved confusion matrix and ROC curve for the retained test at threshold 0.5. The selected model has ROC-AUC 0.9240.

C. Ablation and uncertainty

A development-only ablation holds regularization at 0.01 and uses the same folds. All 149 features reach 95.38% mean balanced accuracy, whereas the 21 histogram/global summaries reach 88.12%. The 7.25 percentage-point difference supports the value of coarse spatial summaries under this protocol. It is not an independent confirmation on unseen external data, and the ablation does not alter the final model after test evaluation.

TABLE 5. Grouped percentile 95% intervals, conditional on the fitted model.

Test measure	95% interval
Balanced accuracy	75.71%-82.39%
Recall	94.67%-98.26%
Specificity	54.92%-67.82%
F1	0.8516-0.9058
ROC-AUC	0.8990-0.9436

VII. DISCUSSION AND LIMITATIONS

The development comparison favors a discriminative linear model over the selected generative baseline for these features. That finding is specific to this dataset, partition policy and small candidate set. It does not demonstrate superiority over richer classical methods or neural models that were not evaluated. Likewise, the ablation shows a development benefit from spatial information without identifying its clinical cause.

The CV-to-test gap is a central result. Distribution differences between the supplied splits, limited feature expressiveness and acquisition shortcuts are plausible explanations, but this experiment does not isolate their causal contributions. Proxy grouping and exact deduplication remove known overlap while leaving uncertainty about true patient identity and near duplicates.

The coarse representation can discard subtle findings and encode exposure, padding or markers. Disease-specific interpretation is not established. Dataset annotations are inherited and not independently adjudicated. The test remains internal to one public dataset, so performance across hospitals, age ranges or acquisition protocols is unknown. Weighted logistic scores and equal-prior Naive Bayes outputs require separate calibration analysis before any probability interpretation.

The bootstrap intervals describe only resampling variation within the retained test proxy groups. They do not incorporate retraining, candidate selection or external shifts. Future work should prioritize verified patient/site metadata and external validation, followed by controlled representation comparisons and calibration analysis. These are proposed experiments, not completed results. The present work supports education and reproducible benchmarking, not clinical deployment.

VIII. REPRODUCIBILITY

The standalone notebook is named Chest X-Rays Pneumonia.ipynb. Local execution uses Python 3.12.6, NumPy 2.4.2, Pillow 12.2.0 and Matplotlib 3.10.8. In Colab, a CPU runtime suffices. The data cell first checks image locations and attempts a public Kaggle archive download if images are absent. If access fails, the user must upload and extract the dataset at the indicated path. No credential or raw radiograph is included in the repository.

Outputs under basic_phase1 include the image audit, group-fold assignments, candidate comparison, ablation, test predictions, numerical model parameters, figures and final evaluation JSON. A local feature cache supports verification after rerunning the notebook and remains excluded from Git. The verification script checks finite-difference logistic gradients, perfect/reversed/tied AUC cases, constant-column scaling, decreasing optimization loss, disjoint development/test groups, single-fold group membership and agreement after parameter reloading. Saved notebook execution is local; no Colab training execution is claimed.

IX. CONCLUSION

An explicit 149-feature pipeline selects regularized logistic regression using grouped development validation. The retained internal test yields 83.50% accuracy, 96.64% recall and 61.47% specificity at the fixed threshold 0.5. Spatial summaries improve the controlled development comparison, but modest specificity and a 16.32 percentage-point CV-to-test balanced-accuracy gap constrain generalization. The main outcome is a transparent, reproducible baseline with documented assumptions and errors.

REFERENCES

[1] D. S. Kermany et al., “Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning,” Cell, vol. 172, no. 5, pp. 1122-1131.e9, 2018. doi:10.1016/j.cell.2018.02.010.

[2] A. Y. Ng and M. I. Jordan, “On Discriminative vs. Generative Classifiers: A comparison of logistic regression and naive Bayes,” Advances in Neural Information Processing Systems, vol. 14, 2001.

[3] T. Hastie, R. Tibshirani and J. Friedman, The Elements of Statistical Learning, 2nd ed. New York, NY, USA: Springer, 2009. doi:10.1007/978-0-387-84858-7.

[4] J. R. Zech, M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano and E. K. Oermann, “Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study,” PLOS Medicine, vol. 15, no. 11, e1002683, 2018. doi:10.1371/journal.pmed.1002683.

[5] P. Mooney, “Chest X-Ray Images (Pneumonia),” Kaggle dataset. https ://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia (accessed Oct. 27, 2026).